Extended fitting methods for multi-angle, polarized photon correlation spectroscopy: generalizing and relaxing the assumptions of CORENN

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Abstract

The CORENN algorithm recovers a particle-size distribution (PSD) from a single dynamic-light-scattering (DLS) correlation curve by a non-linear, noise-weighted, Tikhonov-regularized, non-negativity-constrained inversion of the Siegert model. It is, however, built on four assumptions that are violated by the photon-correlation-tomography (PCT) measurements we wish to interpret: (i) a single scattering angle, so that the intensity weighting of each size class is treated as q-independent; (ii) optically isotropic, spherical scatterers undergoing purely *translational* diffusion; (iii) a correlation curve supplied entirely by the hardware correlator; and (iv) a parametric noise model in place of measured replicate statistics. This document develops a unified inverse-problem framework that relaxes each assumption. We (1) promote the unknown to a q-consistent distribution and fold the angular dependence of the scattering through the particle form factor $P(qR)$; (2) add the *rotational* degree of freedom probed by depolarized (VH) scattering, yielding a joint distribution over translational and rotational dynamics $f(D_\mathrm{t}, D_\mathrm{r})$ from which size *and* shape/anisotropy are recovered; (3) extend the usable lag range by a graceful, uncertainty-weighted merge of the hardware correlogram with a software correlation of the photon count-rate trace; and (4) replace the noise model with a replicate-calibrated, Schätzel-shaped weighting. The resulting estimator is a single regularized non-linear weighted least-squares problem over a non-negative distribution, of which standard CORENN is the spherical, single-angle, translational-only special case.

1 Background: the CORENN forward model

The measured intensity correlation function (ICF) is related to the field correlation function (FCF) by the Siegert relation

$$g^{(2)}(q,\tau) - 1 = \beta \left| g^{(1)}(q,\tau) \right|^2, \tag{1}$$

with coherence factor $\beta \leq 1$. For a dilute, ergodic suspension of spherical scatterers the FCF is the Laplace transform of the intensity-weighted distribution of decay rates,

$$g^{(1)}(q,\tau) = \int_0^\infty I(R)\, e^{-q^2 D(R)\, \tau}\, \mathrm{d}R, \qquad D(R) = \frac{k_\mathrm{B} T}{6\pi\eta R}, \qquad \int_0^\infty I(R)\, \mathrm{d}R = 1, \tag{2}$$

where $I(R)$ is the fraction of scattered intensity arising from particles of hydrodynamic radius R, $D(R)$ is the Stokes–Einstein translational diffusion coefficient, η the solvent viscosity and $q = (4\pi n/\lambda_0)\sin(\theta/2)$ the scattering-vector magnitude. CORENN discretizes $I(R)$ on a cubic B-spline basis $\{\Phi_j\}$,

$$I(R) \approx \sum_{j=1}^{m} x_j\, \Phi_j(R), \qquad x_j \geq 0, \tag{3}$$

so that, defining the kernel $K_{ij} = \int \Phi_j(R)\, e^{-q^2 D(R)\tau_i}\, \mathrm{d}R$, the model reads $g^{(2)}(\tau_i) - 1 = \beta\,(K\mathbf{x})_i^2$. The PSD is then the solution of the regularized non-linear weighted least-squares (NLWLS) problem

$$\hat{\mathbf{x}}(\lambda) = \arg\min_{\mathbf{x}\geq 0} \sum_i w_i^2 \big[\, \hat{g}_i - \beta\,(K\mathbf{x})_i^2 - b \,\big]^2 \;+\; \lambda\, \|L\mathbf{x}\|^2, \tag{4}$$

with data weights $w_i = 1/\sigma_i$, a free baseline b, a curvature penalty L (the second-derivative operator $\int [I''(R)]^2\, \mathrm{d}R = \mathbf{x}^\top L^\top L\, \mathbf{x}$), and the regularization weight λ chosen by generalized cross validation (GCV). The defining feature relative to CONTIN is that the *square* in Eq. (**??**) is retained: one does not linearize by taking $\sqrt{(\hat{g}-1)/\beta} = K\mathbf{x}$, which would distort the (heteroscedastic) noise.

Assumptions to be relaxed. Equations (**??**)–(**??**) embed: a single q with q-independent weighting $I(R)$ (§**??**); spherical, optically isotropic particles with only translational diffusion (§**??**); a hardware-supplied ICF over a fixed lag window (§**??**); and a parametric noise model for w_i (§**??**).

2 Multi-angle inversion with explicit form-factor weighting

The intensity scattered by a particle of radius R at scattering vector q is not q-independent: for a homogeneous sphere in the Rayleigh–Gans–Debye limit it scales as $R^6 P(qR)$, with the form factor

$$P(qR) = \left[\, \frac{3\big(\sin qR - qR\cos qR\big)}{(qR)^3} \,\right]^2, \qquad P(qR) \to 1 \text{ as } qR \to 0. \tag{5}$$

Writing the number distribution as $N(R)$, the intensity weighting at angle q is $I(R,q) \propto N(R)\, R^6\, P(qR)$. The crucial observation is that the R^6 factor is q-independent, so we may define the *angle-extrapolated intensity distribution*

$$I_0(R) \;\equiv\; \frac{N(R)\, R^6}{\int N(R')\, R'^6\, \mathrm{d}R'}, \tag{6}$$

i.e. the intensity PSD one would measure as $q \to 0$. The per-angle FCF is then

$$g^{(1)}(q,\tau) = \frac{\displaystyle\int I_0(R)\, P(qR)\, e^{-q^2 D(R)\,\tau}\, \mathrm{d}R}{\displaystyle\int I_0(R)\, P(qR)\, \mathrm{d}R}, \tag{7}$$

where the denominator restores $g^{(1)}(q,0) = 1$. This relaxes CORENN's implicit assumption that the same $I(R)$ applies at every angle: the only q-dependent re-weighting of a fixed $I_0(R)$ is the bounded, $O(1)$ factor $P(qR)$, which is numerically benign (unlike the $\sim$20-decade dynamic range of R^6 itself).

Discretizing $I_0(R) = \sum_j x_j \Phi_j(R)$ gives, for each angle $q^{(a)}$, a kernel $K_{ij}^{(a)} = \int \Phi_j(R)\, P(q^{(a)}R)\, e^{-(q^{(a)})^2 D(R)\tau_i}\, \mathrm{d}R$ and a model $g_a^{(2)}(\tau_i) - 1 = A_a\,(K^{(a)}\mathbf{x})_i^2$, with a per-angle amplitude A_a (coherence factor divided by the squared normalization) that can be profiled out in closed form,

$$A_a(\mathbf{x}) = \frac{\sum_i w_{a,i}^2\, \hat{g}_{a,i}\, (K^{(a)}\mathbf{x})_i^2}{\sum_i w_{a,i}^2\, (K^{(a)}\mathbf{x})_i^4}. \tag{8}$$

The joint estimator solves a single non-negative, regularized NLWLS over the q-independent coefficients $\mathbf{x}$ using *all* angles simultaneously,

$$\hat{\mathbf{x}}(\lambda) = \arg\min_{\mathbf{x}\geq 0} \sum_a \sum_i w_{a,i}^2 \big[\hat{g}_{a,i} - A_a(\mathbf{x})\,(K^{(a)}\mathbf{x})_i^2\big]^2 + \lambda\,\|L\mathbf{x}\|^2. \tag{9}$$

Because the form factor develops minima and lobes once $qR \gtrsim \pi$, the angular data carry information about the *large*-particle tail that a single back-scattering angle cannot resolve; conversely the small-particle mode, whose intensity weight is suppressed by R^6, remains weakly constrained (an intrinsic DLS limitation, not a defect of the inversion). The number and volume distributions follow from $N(R) \propto I_0(R)/R^6$ and $V(R) \propto I_0(R)/R^3$.

3 Polarized DLS: adding rotational diffusion

For optically anisotropic or non-spherical particles the scattered field has a depolarized component, and the dynamics carry information about rotation. Decompose the scattering into polarized (VV; incident and detected polarizations both vertical) and depolarized (VH; crossed) channels. To leading order in the optical anisotropy, the field correlation functions are

$$g^{(1)}_{\mathrm{VV}}(q,\tau) = \int f(D_\mathrm{t})\, e^{-q^2 D_\mathrm{t}\,\tau}\,\mathrm{d}D_\mathrm{t} \;+\; \tfrac{4}{3}\,\delta^2 \!\int f(D_\mathrm{t}, D_\mathrm{r})\, e^{-(q^2 D_\mathrm{t} + 6D_\mathrm{r})\tau}\,\mathrm{d}D_\mathrm{t}\,\mathrm{d}D_\mathrm{r}, \tag{10}$$

$$g^{(1)}_{\mathrm{VH}}(q,\tau) = \int f(D_\mathrm{t}, D_\mathrm{r})\, e^{-(q^2 D_\mathrm{t} + 6D_\mathrm{r})\,\tau}\,\mathrm{d}D_\mathrm{t}\,\mathrm{d}D_\mathrm{r}, \tag{11}$$

where D_t and D_r are the translational and rotational diffusion coefficients, δ is the (small) optical-anisotropy factor, and $f(D_\mathrm{t}, D_\mathrm{r})$ is the joint distribution of dynamics weighted by scattered intensity. The depolarized channel (**??**) decays with the *enhanced* rate $\Gamma_{\mathrm{VH}} = q^2 D_\mathrm{t} + 6D_\mathrm{r}$: at low q it is dominated by the q-independent rotational term $6D_\mathrm{r}$, providing direct access to rotation. For spheres $\delta = 0$ and the VH channel vanishes, recovering the CORENN picture; the polarized framework is therefore a strict generalization.

From dynamics to geometry. The two diffusion coefficients constrain two geometric parameters. For a spherical equivalent, $D_\mathrm{t} = k_\mathrm{B}T/6\pi\eta R_\mathrm{t}$ and $D_\mathrm{r} = k_\mathrm{B}T/8\pi\eta R_\mathrm{r}^3$ define translational and rotational hydrodynamic radii; their ratio

$$\rho \;\equiv\; \frac{D_\mathrm{r}}{D_\mathrm{t}}\,R_\mathrm{t}^2 \;=\; \frac{3}{4}\left(\frac{R_\mathrm{t}}{R_\mathrm{r}}\right)^3 \tag{12}$$

is a dimensionless shape descriptor (unity for a sphere, > 1 for elongated particles). More generally, for an axisymmetric ellipsoid of semi-axes (a, b) the Perrin friction factors map $(D_\mathrm{t}, D_\mathrm{r})$ to the aspect ratio $p = a/b$ and volume, so the recovered $f(D_\mathrm{t}, D_\mathrm{r})$ becomes a joint distribution over size and shape.

Joint VV/VH multi-angle inverse problem. Combining §**??** and the present section, the unknown is a non-negative joint distribution $f(R,\, s)$ over hydrodynamic size R and a shape coordinate s (e.g. aspect ratio), discretized on a tensor B-spline basis $f(R,s) = \sum_{jk} x_{jk}\,\Phi_j(R)\,\Psi_k(s)$, $x_{jk} \geq 0$. Each (angle, polarization) channel $c = (q^{(a)}, \mathrm{pol})$ contributes a kernel

$$K^{(c)}_{i,jk} = \iint W_c(R, s, q^{(a)})\,\Phi_j(R)\,\Psi_k(s)\,\exp\!\big[-\Gamma_c(R, s, q^{(a)})\,\tau_i\big]\,\mathrm{d}R\,\mathrm{d}s, \tag{13}$$

with channel-specific weight W_c (intensity × form factor × polarization selection rule) and decay rate

$$\Gamma_c = \begin{cases} q^2\, D_{\mathrm{t}}(R) & \text{VV, leading order,} \\ q^2\, D_{\mathrm{t}}(R) + 6\, D_{\mathrm{r}}(R,s) & \text{VH.} \end{cases} \tag{14}$$

All channels are inverted jointly,

$$\hat{\mathbf{x}}(\lambda) = \arg\min_{\mathbf{x}\geq 0} \sum_c \sum_i w_{c,i}^2 \big[\hat{g}_{c,i} - A_c(\mathbf{x})\,(K^{(c)}\mathbf{x})_i^2\big]^2 + \lambda\, \mathbf{x}^\top \Omega\, \mathbf{x}, \tag{15}$$

where Ω is a separable curvature penalty enforcing smoothness in both R and s. Equation (**??**) reduces to standard CORENN (**??**) for a single VV angle and s-independent (spherical) kinematics, and to the multi-angle problem (**??**) when the VH channels and shape coordinate are dropped.

4 Hybrid hardware/software correlation for an extended lag range

Resolving both the fast rotational dynamics of §**??** and the slow dynamics of large aggregates requires a correlation function spanning many more decades in τ than any single estimator provides. The hardware correlator is accurate from ∼ns to the end of the main relaxation but, at long lags, suffers triangular-averaging distortion, few independent decorrelation windows, and irreproducible slow "artifacts." Meanwhile the instrument streams a photon count-rate trace $n(t)$ binned every few milliseconds for the whole acquisition. Software-correlating that trace — and ensemble-averaging across replicates — yields a superior estimate of the slow dynamics. We use the symmetric-normalization, cross-correlation estimator

$$g_{\mathrm{sw}}^{(2)}(\tau_k) = \frac{\big\langle n_A(t_i)\, n_B(t_i+\tau_k)\big\rangle_i}{\big\langle n_A(t_i)\big\rangle_i \big\langle n_B(t_i+\tau_k)\big\rangle_i}, \tag{16}$$

with the direct and delayed monitor averages taken over the same overlapping segment (suppressing drift bias) and channels $A \times B$ cross-correlated to remove detector afterpulsing and dead-time.

Graceful, uncertainty-weighted merge. Rather than hard-truncating the hardware curve at an arbitrary lag, the two estimators are blended by inverse-variance weighting on a common log-lag grid:

$$g_{\mathrm{merge}}^{(2)}(\tau) - 1 = \frac{\sigma_{\mathrm{hw}}^{-2}\,[g_{\mathrm{hw}}^{(2)} - 1] + \sigma_{\mathrm{sw}}^{-2}\,[g_{\mathrm{sw}}^{(2)} - 1]}{\sigma_{\mathrm{hw}}^{-2} + \sigma_{\mathrm{sw}}^{-2}}, \qquad \sigma_{\mathrm{merge}}^{-2} = \sigma_{\mathrm{hw}}^{-2} + \sigma_{\mathrm{sw}}^{-2}. \tag{17}$$

The software amplitude is first matched to the hardware over a narrow window straddling the handoff (both measure $\beta\,|g^{(1)}|^2$ for the same field, but with different coherence factors). The expected uncertainties encode where each estimator is trustworthy:

$$\sigma_{\mathrm{hw}}(\tau) = \sigma_{\mathrm{hw}}^{\mathrm{base}}(\tau)\,\Big[1 + (\tau/\tau_{\mathrm{distort}})^p\Big], \tag{18}$$

$$\sigma_{\mathrm{sw}}(\tau) = \frac{\sigma_{\mathrm{sw}}^{\mathrm{base}}(\tau)}{\sqrt{N_{\mathrm{rep}}}}\,\Big[1 + (\,m\,\Delta t_{\mathrm{bin}}/\tau)^p\Big], \tag{19}$$

so the hardware is down-weighted only beyond the distortion lag τ_{distort}, while the software is down-weighted within a few bin times $m\,\Delta t_{\mathrm{bin}}$ of its shortest lag. The crossover then falls naturally just after the main decay, and $\sigma_{\mathrm{merge}}(\tau)$ is exactly the weight $w_i = 1/\sigma_i$ required by the inversions of §§**??**–**??**.

5 Noise model: replicate-calibrated Schätzel weighting

The statistical noise on the ICF estimator grows at long lag because the number of independent decorrelation windows in a run of duration T falls as $M(\tau) \sim T/\tau$ (Schätzel 1990; Biganzoli & Ferri 2018). A robust, numerically stable model for the variance of $g^{(2)} - 1$ is

$$\sigma^2(\tau) = \sigma_{\text{floor}}^2 + g_{\text{a}} \frac{\left[1 + \beta\, g^{(1)}(\tau)^2\right]^2}{M(\tau)}, \qquad M(\tau) = \max\big(T/\tau,\, 1\big), \tag{20}$$

where σ_{floor} is the short-lag (shot/quantization) level and $g_{\text{a}} = O(1)$ a self-beating gain. When replicate curves are available (the PCT data provide five per measurement), we take the *shape* of $\sigma(\tau)$ from Eq. (**??**) but calibrate its absolute level to the measured short-lag replicate scatter, and never under-weight a lag the replicates flag as noisy: $\sigma \leftarrow \max(\sigma_{\text{model}}, \sigma_{\text{empirical}})$. This combines the long-lag growth that the hardware artifacts demand with the empirical fidelity of direct replicate statistics, and is what allows the inversion to ignore the long-lag noise floor rather than fit it with a spurious slow population.

6 Summary of the generalization

Table **??** situates the four extensions relative to standard CORENN. Each is implemented in the accompanying `corenn.py` (inversion, form-factor multi-angle, replicate/Schätzel weighting, uncertainty bands) and `hybrid_correlator.py` (software correlation and graceful merge); the polarized joint inversion (**??**) is the natural next step within the same NLWLS machinery.

Assumption in CORENN	Generalization	Mechanism
Single angle, q-indep. $I(R)$	Multi-angle $I_0(R)$	form factor $P(qR)$, Eq. (**??**)
Spherical, isotropic, transl. only	Rotational dynamics $f(D_{\text{t}}, D_{\text{r}})$	VV/VH channels, Eqs. (**??**)–(**??**)
Hardware ICF, fixed lag window	Hybrid hardware+software	inverse-variance merge, Eq. (**??**)
Parametric noise model	Replicate-calibrated weights	Schätzel shape, Eq. (**??**)

Table 1: Each row relaxes one CORENN assumption while preserving the non-negative, regularized, non-linear weighted least-squares structure; CORENN is the common special case.

Relation to existing PCT theory. This document extends the basic FCF/ICF and Siegert treatment of `dls_theory.tex`; Eqs. (**??**)–(**??**) here are the polydisperse generalization of its Eq. (13), and §**??** supplies the rotational content absent from the spherical, translational-only model used there.